

A Finite-Cutoff Hilbert-Space Model

for Prime–Zero Energy Structure

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Abstract

We construct a finite-cutoff Hilbert-space model for the spectral structure associated with primes and nontrivial zeros of the Riemann zeta function. Given a prime cutoff κ and truncated zero ordinates $\{\gamma_1, \dots, \gamma_N\}$, we define two finite-dimensional real Hilbert spaces H_{str} (over primes) and H_{null} (over zeros), connected by a linear map $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$, and the self-adjoint loop operator $T = \Phi^* \circ \Phi$.

What is proved. We establish the algebraic identity $\Delta := E_{\text{str}} - E_{\text{spec}} = \sum_p c_p^2 \|a_p\|^2 - \|\sum_p c_p a_p\|^2$ by direct expansion.

What is numerical. $\Delta > 0$ and $\eta_{\text{orig}} = \Delta/E_{\text{str}} \in [0.598, 0.704]$ for benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$.

What remains open. An analytic proof of $\eta_{\text{orig}} > 0$ remains open.

Structure. The formal core in §§ 2–5 is non-circular and proof-complete; results from Papers 1–2 appear only as context. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–5. The model suggests a picture in which $\sigma = \frac{1}{2}$ plays the role of an energy reference point.

Notation and Conventions

Critical line. Throughout this series, σ denotes $\Re(s)$ and the critical line means $\sigma = \frac{1}{2}$. The functional equation symmetry $\xi(s) = \xi(1-s)$ makes $\sigma = \frac{1}{2}$ the natural reference point.

Global parameters. In this paper, $\kappa = 53$ is the prime cutoff (16 active primes $p \leq \kappa$), $\varepsilon = 0.05$ the Gaussian regularization parameter, and $N = 100$ the number of Riemann zero ordinates $0 < \gamma_1 \leq \gamma_2 \leq \dots \leq \gamma_N$ employed.

Main objects. The prime space $H_{\text{str}} = \ell^2(S(\kappa))$ and zero space $H_{\text{null}} = \ell^2(\{\gamma_1, \dots, \gamma_N\})$ are connected by $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$, $\Phi(e_p) = a_p$, where $(a_p)_k = e^{-\varepsilon^2 \gamma_k^2/2} \sin(\gamma_k \log p)$. The loop operator is $T = \Phi^* \circ \Phi$ and $\eta_{\text{orig}} = 1 - E_{\text{spec}}/E_{\text{str}}$ measures the energy gap.

Weights. The renormalized weight squared

$$f_p := V_p(\tfrac{1}{2}) - \frac{4(\log p)^2}{p} = \frac{4(\log p)^2(2p-1)}{p(p-1)^2}$$

subtracts the leading divergent term from the local prime curvature $V_p(\frac{1}{2})$ of Paper 1. The canonical weight in the η -framework of this paper is

$$c_p^\eta := \sqrt{f_p}.$$

Not to be confused with. $T = \Phi^*\Phi$ (loop operator on H_{str} , this paper) is distinct from $\tilde{T} = \Phi\Phi^*$ (the adjoint loop operator on H_{null}). $D \approx 9.471$ (renormalized energy sum, Paper 2) is distinct from $E_{\text{str}} \approx 6.119$ (structural energy at benchmark parameters, this paper). The canonical weight $c_p^\eta = \sqrt{f_p}$ in the η -framework of this paper has the same numerical value as Paper 2's renormalized weight $c_p^{\text{ren}} = \sqrt{f_p}$ but plays a distinct role: c_p^η enters $E_{\text{str}} = \sum_p (c_p^\eta)^2 \|a_p\|^2$ inside the η -formula, whereas c_p^{ren} enters $D = \sum_p (c_p^{\text{ren}})^2$ in the Weil renormalization.

1 Introduction

1.1 Background and Motivation

The present paper is the third in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The formal core in §§ 2–4 is non-circular and proof-complete.

Paper 1 [6] introduced the curvature function $H_\xi(\sigma, t) := \frac{d^2}{d\sigma^2} \log |\xi(\sigma + it)|^2$ and established the divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \rightarrow \infty$ at $\sigma = \frac{1}{2}$, contrasted with convergence for $\sigma > \frac{1}{2}$.

Paper 2 [7] connected this local curvature to the Weil explicit functional, constructing admissible test functions and establishing the convergent weight sum $D = \sum_p (c_p^{\text{ren}})^2 \approx 9.471$. These results appear in § 2 for motivation and context; the formal core does not depend on them.

The present paper makes the geometric structure underlying both papers explicit: two finite-dimensional Hilbert spaces H_{str} and H_{null} , connected by a linear map Φ , and a self-adjoint loop operator $T = \Phi^* \circ \Phi$.

Several established programs seek a spectral interpretation of the Riemann zeros. Berry and Keating [1] studied a semiclassical analogue via quantum chaos; Connes [2] realizes the zeros through a trace formula in noncommutative geometry; de Branges [3] approached the problem through Hilbert spaces of entire functions. The present model differs from all three: it constructs *two* finite-dimensional spaces coupled by a bilinear map, rather than seeking a single operator whose spectrum is $\{\gamma_k\}$. The Gram-matrix construction $T_{pq} = \langle a_p, a_q \rangle$ from damped sine vectors appears to be new; conceptually it may be viewed as a discrete analogue of spectral interpretations of the explicit formula, as studied by Goldfeld [4]. The test functions of Paper 2 connect to adelic Schwartz functions as in Tate’s thesis [9]; the Weil positivity criterion [10, 5] provides the outer framework.

The renormalized framework of Paper 2 and the η -framework of this paper use different normalizations; a precise identification remains open (Open Problem 5.5).

1.2 Main Results

We define two finite-dimensional real Hilbert spaces H_{str} (over primes $p \leq \kappa$) and H_{null} (over zero ordinates $\gamma_1, \dots, \gamma_N$), connected by the linear map $\Phi : H_{\text{str}} \rightarrow H_{\text{null}}$ built from Gaussian-damped prime resonance vectors (§ 3). The self-adjoint loop operator $T = \Phi^* \circ \Phi$ is the Gram matrix of these vectors (§ 3).

We prove the algebraic identity $\Delta := E_{\text{str}} - E_{\text{spec}}$ by direct expansion (Theorem 4.3). Numerically, $\eta_{\text{orig}} = \Delta/E_{\text{str}} \in [0.598, 0.704]$ for benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$ (Observation 4.4). An analytic proof of $\eta_{\text{orig}} > 0$ remains open (Open Problem 5.1).

1.3 Reader Contract

This paper *proves*, by direct algebraic expansion, the variance identity $\Delta = E_{\text{str}} - E_{\text{spec}}$ (Theorem 4.3) and the basic properties of the loop operator $T = \Phi^* \circ \Phi$: self-adjointness, positive semi-definiteness, and σ -invariance of the normalized Gram matrix (Proposition 3.5). These are unconditional.

This paper *establishes numerically* that $\eta_{\text{orig}} > 0$ for all tested parameter combinations, with $\eta_{\text{orig}} \in [0.598, 0.704]$ at benchmark parameters (§5). The uniform spectral bound $B_{\text{max}} \leq 0.106 < 1$ and its σ -invariance are numerical observations.

This paper *leaves open* the analytic proof of $\eta_{\text{orig}} > 0$ (Problem 6.1), the spectral bound for canonical weights (Problem 6.2), the analytical cancellation bound (Problem 6.3), the σ -invariance of B_{max} (Problem 6.4), and the renormalized shell energy (Problem 6.5).

2 Two Hilbert Spaces

We work throughout with real Hilbert spaces; all vectors and operators have real entries. Complex phasor language appears in §3.1 as heuristic motivation only; all formal objects ($H_{\text{str}}, H_{\text{null}}, \Phi, T$) are real.

Throughout this paper, $\kappa > 1$ is a fixed real cutoff parameter, $S(\kappa) := \{p \text{ prime} : p \leq \kappa\}$ the set of active primes, and $\{\gamma_k\}_{k=1}^N$ the first N positive imaginary parts of nontrivial zeros of $\zeta(s)$, ordered $0 < \gamma_1 \leq \gamma_2 \leq \dots$. The Gaussian damping parameter $\varepsilon > 0$ is fixed throughout.

Definition 2.1 (Prime space H_{str}).

$$H_{\text{str}} := \ell^2(S(\kappa))$$

with basis $\{e_p : p \in S(\kappa)\}$, standard inner product $\langle a, b \rangle_{\text{str}} = \sum_p a_p b_p$, and $\dim H_{\text{str}} = |S(\kappa)| = \pi(\kappa)$. The structural energy of a weight vector $c \in H_{\text{str}}$ is $E_{\text{str}}(c) = \sum_p c_p^2 \|a_p\|^2$ (defined in §4 after Φ is introduced).

Definition 2.2 (Zero space H_{null}).

$$H_{\text{null}} := \ell^2(\{\gamma_1, \dots, \gamma_N\})$$

with basis $\{f_k : k = 1, \dots, N\}$, inner product $\langle u, v \rangle_{\text{null}} = \sum_{k=1}^N u_k v_k$ (unweighted), and $\dim H_{\text{null}} = N$.

Remark (Why $\sigma = \frac{1}{2}$ is a distinguished reference point). This remark provides context; it is not used in §§3–5.

Two independent proven reasons support this choice:

(i) Symmetry [proven]: $\sigma = \frac{1}{2}$ is the unique fixed point of $s \mapsto 1 - s$, the symmetry of the completed zeta function $\xi(s) = \xi(1 - s)$.

(ii) Curvature singularity [proven, Papers 1–2]: $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \rightarrow \infty$ as $\kappa \rightarrow \infty$, while $H_{\text{local}}(\sigma, \kappa) \rightarrow C(\sigma) < \infty$ for all $\sigma > \frac{1}{2}$.

$\sigma = \frac{1}{2}$ is thus a distinguished reference point by symmetry and curvature divergence. Energy near-equality ($E_{\text{str}} \approx E_{\text{spec}}$) is NOT observed in the finite-cutoff model (see §5 and §8).

3 The Linear Map Φ and the Loop Operator T

Convention. Gaussian damping is carried by the vectors: $a_p = (e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p))_k$. The H_{null} inner product is unweighted: $\langle u, v \rangle = \sum_k u_k v_k$.

3.1 Motivation for the resonance vectors

The formal definition is Definition 3.2 below; this subsection provides intuition.

The prime resonance vector at zero ordinate γ_k and prime p arises as the imaginary part of the prime phasor $\varphi_p(u) = e^{iu \log p}$ evaluated at $u = \gamma_k$, with Gaussian regularization for convergence:

$$(a_p)_k = e^{-\varepsilon^2 \gamma_k^2 / 2} \cdot \sin(\gamma_k \log p).$$

The k -th component measures how strongly prime p resonates with zero ordinate γ_k .

Definition 3.1 (Prime resonance vectors). *For each prime $p \in S(\kappa)$, the canonical resonance vector at $\sigma = \frac{1}{2}$ is*

$$a_p^{\text{can}} := a_p := (e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p))_{k=1}^N \in H_{\text{null}}.$$

At general σ , the resonance vectors depend on σ via

$$a_p^{(\sigma)} := p^{-(\sigma-1/2)} \cdot a_p^{\text{can}}.$$

The vectors $a_p = a_p^{\text{can}} = a_p^{(1/2)}$ at $\sigma = \frac{1}{2}$ are the canonical representatives used in all primary definitions.

Definition 3.2 (Linear map Φ).

$$\Phi: H_{\text{str}} \rightarrow H_{\text{null}}, \quad \Phi(e_p) := a_p^{\text{can}}.$$

Extended linearly: $\Phi(c) = \sum_p c_p a_p^{\text{can}}$ for any $c = \sum_p c_p e_p \in H_{\text{str}}$.

We define Φ as a linear map; no injectivity claim is made for the finite-cutoff model.

Definition 3.3 (Gram matrices). *The unnormalized Gram matrix (canonical) is*

$$G_{pq}^{\text{un}} := G_{pq}^{\text{un,can}} := \langle a_p^{\text{can}}, a_q^{\text{can}} \rangle_{H_{\text{null}}} = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \sin(\gamma_k \log p) \sin(\gamma_k \log q).$$

The σ -dependent extension is $G^{\text{un}}(\sigma)_{pq} = p^{-(\sigma-1/2)} q^{-(\sigma-1/2)} G_{pq}^{\text{un}}$. The normalized Gram matrix

is

$$G_{pq}^{\text{norm}} := \frac{G_{pq}^{\text{un}}}{\|a_p^{\text{can}}\| \cdot \|a_q^{\text{can}}\|} = \left\langle \frac{a_p^{\text{can}}}{\|a_p^{\text{can}}\|}, \frac{a_q^{\text{can}}}{\|a_q^{\text{can}}\|} \right\rangle_{H_{\text{null}}}.$$

The matrix G^{norm} is σ -invariant (Proposition 3.5(iv)).

Definition 3.4 (Loop operator T).

$$T := \Phi^* \circ \Phi: H_{\text{str}} \longrightarrow H_{\text{str}}.$$

The matrix representation of T in the prime basis is $T_{pq} = G_{pq}^{\text{un}}$. T is thus identical to the canonical Gram matrix G^{un} , and is explicitly constructed from prime resonance vectors.

Proposition 3.5 (Properties of T ; proved). (i) Self-adjoint: $T^* = (\Phi^* \Phi)^* = \Phi^* \Phi = T$.

(ii) Positive semi-definite: $\langle Tc, c \rangle = \|\Phi c\|^2 \geq 0$ for all $c \in H_{\text{str}}$.

(iii) Non-negative eigenvalues: All eigenvalues $\lambda_j \geq 0$.

(iv) σ -invariance of G^{norm} [by direct calculation]: For the σ -dependent family $a_p^{(\sigma)}$, the normalized Gram matrix G_{pq}^{norm} is independent of σ . Proof. The scalar factor $p^{-(\sigma-1/2)}$ cancels in the normalized ratio. $\square$

Remark (Spectral structure of T). The eigenvalues $\lambda_j \geq 0$ of $T = \Phi^* \circ \Phi$ are the squares of the singular values $\sigma_j(\Phi)$: $\lambda_j = \sigma_j(\Phi)^2$. The eigenvalues λ_j are built from the zero ordinates γ_k appearing in the definition of a_p^{can} , but are not directly the zeros γ_k themselves. This distinguishes T from the classical Hilbert-Pólya ansatz: T is explicitly constructed, not postulated.

4 The Variance Identity and η_{orig}

Convention. Objects without σ -argument refer to the canonical ($\sigma = \frac{1}{2}$) version throughout § 4.

Definition 4.1 (Canonical and σ -dependent energies). For a weight vector $c \in H_{\text{str}}$ with components c_p :

Canonical quantities (at $\sigma = \frac{1}{2}$):

$$D^{\text{can}} := \text{diag}(\|a_p^{\text{can}}\|^2)_{p \in S(\kappa)}, \quad E_{\text{str}}^{\text{can}} := c^\top D^{\text{can}} c = \sum_p c_p^2 \|a_p^{\text{can}}\|^2,$$

$$E_{\text{spec}}^{\text{can}} := \|\Phi c\|^2 = c^\top G^{\text{un}} c.$$

σ -dependent extensions:

$$D(\sigma)_{pp} := p^{-2(\sigma-1/2)} \cdot D_{pp}^{\text{can}}, \quad E_{\text{str}}(\sigma) := \sum_p c_p^2 p^{-2(\sigma-1/2)} \|a_p^{\text{can}}\|^2,$$

$$E_{\text{spec}}(\sigma) := c^\top G^{\text{un}}(\sigma) c.$$

The canonical weight vector in the η -context is

$$c_p^\eta := \sqrt{f_p} = \sqrt{\frac{4(\log p)^2(2p-1)}{p(p-1)^2}}, \quad p \in S(\kappa).$$

Remark (Canonical weights — normative form). The canonical weight $c_p^\eta = \sqrt{f_p}$ is the normative form used throughout this series. For large p , $c_p^\eta \approx \sqrt{8} \cdot (\log p)/p$; the two forms agree asymptotically but differ for small primes. The numerical results in §5 use the normative form.

Assumption 4.2 (Positive definiteness of D). Throughout this paper we assume $D_{pp} = \|a_p\|^2 > 0$ for all $p \in S(\kappa)$. This holds in all numerical experiments conducted here.

Theorem 4.3 (Algebraic identity; proved). The following equality holds by direct expansion:

$$\Delta := E_{\text{str}} - E_{\text{spec}} = \sum_p c_p^2 \|a_p\|^2 - \left\| \sum_p c_p a_p \right\|^2.$$

Proof. Expand $E_{\text{spec}} = \|\Phi c\|^2 = \langle \sum_p c_p a_p, \sum_q c_q a_q \rangle = \sum_{p,q} c_p c_q \langle a_p, a_q \rangle = c^\top G^{\text{un}} c$. Then $\Delta = c^\top D^{\text{can}} c - c^\top G^{\text{un}} c$, which is the stated identity. $\square$

Observation 4.4 (Non-negativity, numerical).

$$\Delta \geq 0 \quad \text{equivalently} \quad \eta_{\text{orig}} \geq 0$$

holds for all tested parameters with canonical weights $c_p^\eta = \sqrt{f_p}$ (Definition 4.1). [Numerical only — no unconditional proof for general c .]

Note: $\eta_{\text{orig}} \geq 0$ for general weight vector c is equivalent to $\lambda_{\max}(D^{-1/2} T D^{-1/2}) \leq 1$, which is the content of Open Problem 5.2. Numerical experiments show $\lambda_{\max}(D^{-1/2} T D^{-1/2}) \approx 0.39 \cdot \pi(\kappa) \rightarrow \infty$, so the universal bound $\lambda_{\max} \leq 1$ does not hold in general.

Cauchy-Schwarz does **not** guarantee $\eta_{\text{orig}} \geq 0$: the correct identity is $\Delta = c^\top (D^{\text{can}} - G^{\text{un}}) c$; the sign of Δ is not determined by the algebraic identity alone.

Definition 4.5 (η_{orig}).

$$\eta_{\text{orig}} := 1 - \frac{E_{\text{spec}}}{E_{\text{str}}} = 1 - \frac{\|\Phi c\|^2}{E_{\text{str}}}.$$

Boundary cases: $\eta_{\text{orig}} = 0$ iff $E_{\text{spec}} = E_{\text{str}}$; $\eta_{\text{orig}} = 1$ iff $\Phi c = 0$.

Remark (σ -dependence via rescaled weights). The entire σ -dependence of $\eta_{\text{orig}}(\sigma)$ is carried by the rescaled weight vector $\tilde{c}_p(\sigma) := c_p \cdot p^{-(\sigma-1/2)}$:

$$\eta_{\text{orig}}(\sigma) = 1 - \frac{\tilde{c}(\sigma)^\top G^{\text{un}} \tilde{c}(\sigma)}{\tilde{c}(\sigma)^\top D^{\text{can}} \tilde{c}(\sigma)}.$$

Proof. Since $G^{\text{un}}(\sigma)_{pq} = p^{-(\sigma-1/2)} q^{-(\sigma-1/2)} G_{pq}^{\text{un}}$, one has $c^\top G^{\text{un}}(\sigma) c = \tilde{c}^\top G^{\text{un}} \tilde{c}$ and $E_{\text{str}}(\sigma) = \tilde{c}^\top D^{\text{can}} \tilde{c}$. $\square$

Numerical evidence. All computations use $N = 100$ zero ordinates at the reference parameters; the algebraic identity (Theorem 4.3) is verified to rounding error $< 10^{-14}$. Verification scripts are available in the companion repository [8].

Observation 4.6 (η_{orig} positive across all tested parameters). $\eta_{\text{orig}} > 0$ in all tested parameter combinations. For the benchmark slice $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$:

$$\eta_{\text{orig}} \in [0.598, 0.704].$$

The function $\sigma \mapsto \eta_{\text{orig}}(\sigma)$ is strictly monotone decreasing. Note: the range $[0.598, 0.704]$ is strongly ε -dependent; no universal range over all (κ, ε) is claimed.

The following values illustrate the strict monotone decrease of $\eta_{\text{orig}}(\sigma)$ at the reference parameters.

σ	η_{orig}
0.10	0.70441977
0.20	0.69826865
0.30	0.69038398
0.40	0.68072993
0.50	0.66926873
0.60	0.65602069
0.70	0.64110102
0.80	0.62472010
0.90	0.60715692
0.95	0.59802896

Observation 4.7 (Uniform spectral bound $B_{\text{max}} \leq 0.106 < 1$). Define the mode contribution $B_j(\sigma) := \lambda_j(\sigma) |\langle c, u_j(\sigma) \rangle|^2 / E_{\text{str}}(\sigma)$ with consistency identity $\sum_j B_j(\sigma) = 1 - \eta_{\text{orig}}(\sigma)$ [algebraic]. The empirical diagnostic $B_{\text{max}}(\sigma) := \max_{j: \lambda_j > 1} B_j(\sigma)$ satisfies

$$B_{\text{max}}(\kappa, \varepsilon, \sigma) \leq 0.106 < 1$$

for all tested $(\kappa, \varepsilon, \sigma)$.

Selected values:

κ	$\varepsilon = 0.02$	$\varepsilon = 0.05$	$\varepsilon = 0.10$
23	0.1047	0.0220	0.0000
53	0.0793	0.0737	0.0000
101	0.0537	0.0844	0.0324
199	0.0891	0.0461	0.1058

Observation 4.8 ($B_{\text{max}}(\sigma)$ numerically σ -invariant). For $(\kappa, \varepsilon) = (53, 0.05)$, $B_{\text{max}}(\sigma) = 0.0737$ for all tested $\sigma \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$. Analytical explanation: open (Open Problem 5.4).

Remark (Cancellation mechanism, numerical analysis). *The mechanism underlying $B_{\max} < 1$ is a cancellation between the monotone*

weight vector $c_p^\eta = \sqrt{f_p}$ and the oscillatory eigenvectors u_j of G^{un} : for $\kappa = 53$, eigenvectors $u_1, \dots, u_4$ each exhibit 10 sign changes along the prime ordering. Cancellation rates 67–97% observed for $\kappa \in \{23, 53, 101, 199\}$. Formal proof: Open Problem 5.3.

5 Open Problems

Open Problem 5.1 (Analytic positivity). *Prove analytically that $\eta_{\text{orig}} > 0$ for the canonical weight vector $c_p^\eta = \sqrt{f_p}$. Numerically confirmed: $\eta_{\text{orig}} \in [0.598, 0.704]$ for $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$ (§4).*

Open Problem 5.2 (Spectral bound for canonical weights). *Let $T_{\text{ren}} := D^{-1/2}TD^{-1/2}$. Numerical experiments show $\lambda_{\max}(T_{\text{ren}}) \approx 0.39 \cdot \pi(\kappa) \rightarrow \infty$; the universal bound $\lambda_{\max} < 1$ (for all c) does not hold. However, for the canonical weight vector $c_p^\eta = \sqrt{f_p}$, the quantity $\langle T_{\text{ren}}\tilde{c}, \tilde{c} \rangle$ satisfies $\langle T_{\text{ren}}\tilde{c}, \tilde{c} \rangle < 1$ (equivalently $\eta_{\text{orig}} > 0$) for all tested $\kappa \in \{23, 53, 101, 199, 503, 1009\}$. Prove this analytically. The canonical c_p appears orthogonal to the dominant eigenmode of T_{ren} (localized at $\gamma_1 = 14.13\dots$); proving this orthogonality is the core open problem.*

Open Problem 5.3 (Analytical cancellation bound). *Prove analytically:*

$$\max_{j: \lambda_j > 1} \frac{|\langle c, u_j \rangle|^2}{\|c\|^2} \leq C_1 \ll 1 \quad \text{uniformly in } \kappa, j.$$

Approach: Abel summation for $c_p \sim (\log p)/p$ against oscillatory partial sums $\sum_{q \leq p} (u_j)_q$, without reference to RH. Numerically: $C_1 < 0.36$ for $(\kappa, \varepsilon) = (53, 0.05)$.

Open Problem 5.4 (σ -invariance of $B_{\max}$ analytically). *The numerical σ -invariance of $B_{\max}(\sigma)$ (consistent with Proposition 3.5(iv)) is not yet derived analytically from the weight rescaling $c \mapsto \tilde{c}(\sigma)$.*

Open Problem 5.5 (Renormalized shell energy). *The renormalized shell energy $H_{\text{ren}}(\kappa, \varepsilon(\kappa))$ for canonical coupling $N_{\text{eff}} = \pi(\kappa)$ is not yet computed. This lies in the Paper 2 normalization framework.*

6 Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The zero ordinates γ_k enter the construction of Φ as numerically specified inputs, and no property of their distribution requiring the Riemann Hypothesis is assumed or used.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The loop operator $T = \Phi^*\Phi$ is built explicitly from the prime resonance vectors $\{a_p\}_{p \leq \kappa}$. Its construction postulates neither

GUE statistics, Montgomery’s pair correlation conjecture, nor the Hilbert–Pólya conjecture. The eigenvalues λ_j of T are the squares of the singular values of Φ and are not identified with zero ordinates γ_k .

Status separation maintained. The variance identity $\Delta = E_{\text{str}} - E_{\text{spec}}$ is an algebraic theorem; the sign statement $\Delta \geq 0$ is established numerically on the tested grid. These are distinct levels and are flagged separately throughout. No claim relating η_{orig} to RH is proved or used.

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Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo (<https://zenodo.org/search?q=Ulrich+Tehrani>) and GitHub (<https://github.com/utehrani/analysislab-nt>).

Topics of particular interest include: whether $\eta_{\text{orig}} > 0$ is an expected consequence of known equidistribution results for the phases $\{\gamma_k \log p \bmod 2\pi\}$, and whether the observed near-orthogonality of the weight vector to the dominant eigenmode of T connects to standard cancellation phenomena in partial sums involving primes and zeros.

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